BOOKLET A

YISHUN INNOVA JUNIOR COLLEGE
JC 2 PRELIMINARY EXAMINATION
Higher 2

CANDIDATE NAME	

CG		INDEX NO	

BIOLOGY 9744/03

Paper 3 Long Structured and Free-Response Questions **15 September 2021**

2 hours

Candidates answer on the Question Paper.
No Additional Materials are required.

READ THESE INSTRUCTIONS FIRST

Write your name and class in the spaces at the top of this page.
Write in dark blue or black pen on both sides of the paper.
You may use an HB pencil for any diagrams or graphs.
Do not use staples, paper clips, highlighters, glue or correction fluid/tape.

Section A
Answer **all** questions in the spaces provided on the Question Paper.

Section B
Answer any **one** question in the spaces provided on the Question Paper.
Indicate the question you have attempted at the top of page **21**.

The use of an approved scientific calculator is expected, where appropriate.
You may lose marks if you do not show your working or if you do not use appropriate units.

The number of marks is given in brackets [ ] at the end of each question or part question.

At the end of the examination, **submit booklets A, B and C separately** to the invigilator.

For Examiner's Use	
Section A	
1	**/ 27**
2	**/ 13**
3	**/ 10**
Section B	
4 or 5	**/ 25**
Total	**/ 75**

This document consists of **25** printed pages and **3** blank pages.

Section A

Answer **all** the questions in this section.

1 Cystic fibrosis (CF) is one of the most common inherited genetic disease in humans. It is inherited in an autosomal recessive manner.

CF is due to a mutation of the cystic fibrosis transmembrane conductance regulator (CFTR) gene, which codes for CFTR protein. Most CF patients have a mutated CFTR gene on chromosome 7, consisting of deletion of three consecutive nucleotides, resulting in a deletion of the amino acid, phenylalanine, from the amino acid sequence of the CFTR protein.

CFTR proteins function as chloride ion channels embedded in the cell surface membranes of cells that line the respiratory tract, pancreas and intestines. The protein regulates the transport of chloride ions out of these cells.

(a) With reference to the information given above,

(i) explain why cystic fibrosis (CF) is a genetic disease.

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.... [2]

(ii) describe the effect of a mutation of the CFTR gene on the function of the CFTR protein.

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.... [2]

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Almost 40% of adults with CF develop a form of diabetes known as cystic fibrosis-related diabetes (CFRD).

The bacterium *Pseudomonas aeruginosa* can cause chronic (long-lasting) lung infections. A person with CFRD is likely to have poorer lung function and a greater likelihood of having a chronic lung infection than a person who has CF but does not have CFRD.

An investigation was carried out to find out if the severity of damage to lung function in a person with CFRD is affected by

- the gender
- whether or not they have a chronic *P. aeruginosa* infection.

The investigators measured lung function by recording the maximum volume of air that can be expelled from the lungs in the first one second of a forced expiration. This is known as FEV_1. The lower the median FEV_1, the poorer the lung function.

Table 1.1 summarises the results of this investigation. All 812 people in the study had cystic fibrosis.

Table 1.1

	Without chronic *P. aeruginosa* infection				**With chronic *P. aeruginosa* infection**			
	male		**female**		**male**		**female**	
	with CFRD	**without CFRD**	**with CFRD**	**without CFRD**	**with CFRD**	**without CFRD**	**with CFRD**	**without CFRD**
Number of people	44	110	52	93	106	166	121	120
FEV_1	71.1	71.4	53.6	73.6	49.0	59.0	42.0	61.0

(b) With reference to Table 1.1,

(i) discuss whether or not there appears to be a positive correlation between having a chronic *P.aeruginosa* infection and having CFRD.

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.... [2]

(ii) Calculate the percentage difference between the FEV_1 of males and females without CFRD and without *P.aeruginosa* infection.

Show your working.

[2]

The fruit fly, *Drosophila*, has many different species.

Drosophila is commonly used in the laboratories for the study of Type 2 diabetes progression.

Three of these species, *D. pseudoobscura*, *D. persimilis* and *D. Miranda*, are thought to be closely related.

Samples of these three species were collected from the western United States of America. The base sequence of four regions of DNA of each species were sequenced. The divergence of these base sequences in *D. pseudoobscura* and *D. persimilis* from the sequences in *D. miranda* was calculated.

The results are shown in Table 1.2.

Table 1.2

DNA region	*Drosophila* species	Percentage divergence of base sequence from that of *D. miranda*
1	*pseudoobscura*	2.5
	persimilis	2.4
2	*pseudoobscura*	8.1
	persimilis	7.3
3	*pseudoobscura*	2.1
	persimilis	1.7
4	*pseudoobscura*	1.9
	persimilis	1.7

(c) With reference to Table 1.2, describe the evidence that *D. miranda* may be more closely related to *D. persimilis* than to *D. pseudoobscura.*

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.... [2]

(d) Suggest why there is more divergence in some regions of DNA than in others.

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.... [2]

Plasma cells secrete antibody molecules in response to infections caused by bacteria such as *P. aeruginosa*.

Fig. 1.1 is a diagram of an antibody molecule. The diagram is **not** complete.

Fig. 1.1

(e) **(i)** Draw a circle around a variable region. [1]

(ii) Draw in and label the position of the disulfide bonds in the molecule. [1]

(iii) Explain the importance of disulfide bonds in protein molecule, such as antibodies.

[3]

(iv) Describe how antibodies provide protection against bacterium.

[4]

(f) DNA methylation plays an important role in the development of diseases and the process of aging.

In a study, scientists examined DNA methylation at 476,366 sites throughout the genome of plasma cells from a population cohort (N = 421) ranging in age from 14 to 94 years old. Age affects DNA methylation at almost one third of the sites.

Fig. 1.2 shows the effect of DNA methylation with chronological age.

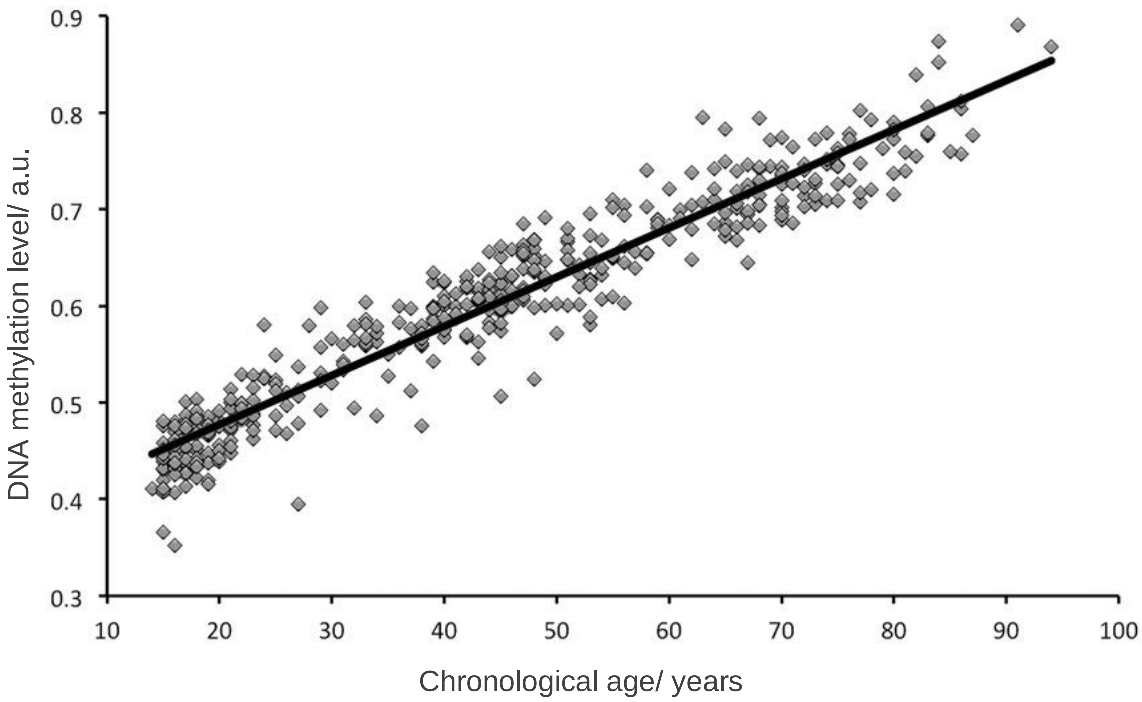

Fig. 1.2

(i) With reference to the Fig. 1.2, discuss the relationship between aging, DNA methylation and immunity.

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.... [4]

(ii) Briefly describe how DNA methylation can lead to cancer.

[2]

[Total: 27]

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BOOKLET B

Name: .... **( )** **CG:** ....

2 In an experiment, genetic material from mouse liver tissues were isolated and treated with DNase 1. DNase 1 is an enzyme that can degrade double-stranded DNA in a non-specific manner. It was found that certain regions of DNA in the cell are naturally protected from degradation by DNase 1.

The DNase 1 enzyme was removed after digestion and any remaining intact DNA samples were extracted and placed under optimal conditions conducive for nucleic acid hybridization to occur.

Radioactively labelled probes complementary to certain genes were then added to the DNA samples. The amount of radioactivity in the samples were determined after excess probes were washed off.

(a) With reference to the principles of nucleic acid hybridization, explain one step that must be taken first before hybridization of the probes can occur.

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.... [2]

The levels of hybridization of labelled probes to two specific gene sequences were measured via radioactivity in arbitrary units (a.u). Some of the results are shown in Table 2.1 below.

Table 2.1

Chromatin sample from liver cells	DNase I treatment	Gene complementary to radioactive probes	Radioactivity (a.u)
1	No	He1	89
2	Yes	He1	27
3	No	Av3	90
4	Yes	Av3	91

(b) With reference to Table 2.1,

(i) identify which gene, *He1* or *Av3*, is transcriptionally active in liver cells.

.... [1]

(ii) explain your answer to **(i)**.

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.... [3]

(c) The concentration and activity level of proteins can be regulated in different cell types through certain mechanisms.

Explain how regulation of either the concentration or activity level of proteins is carried out.

[3]

Cancer is a disease in which normal controls over cell division are lost and malignant tumours form. An early diagnosis of many types of cancer can result in successful treatment.

The BRCA2 protein is involved in suppressing the development of tumours.

Several different dominant alleles of the gene, *BRCA2*, code for faulty versions of the BRCA2 protein. The presence of any one of these faulty alleles leads to an increased chance of developing several types of cancer, including liver cancer. Not everyone with one of these alleles develops cancer. This is because environmental factors, including lifestyle, are also involved.

Fig. 2.1 is a pedigree (family tree) showing the occurrence of cancers in four generations of a family. The presence of a faulty *BRCA2* allele was confirmed in person 15. The other individuals with cancer were not tested for the presence of the allele.

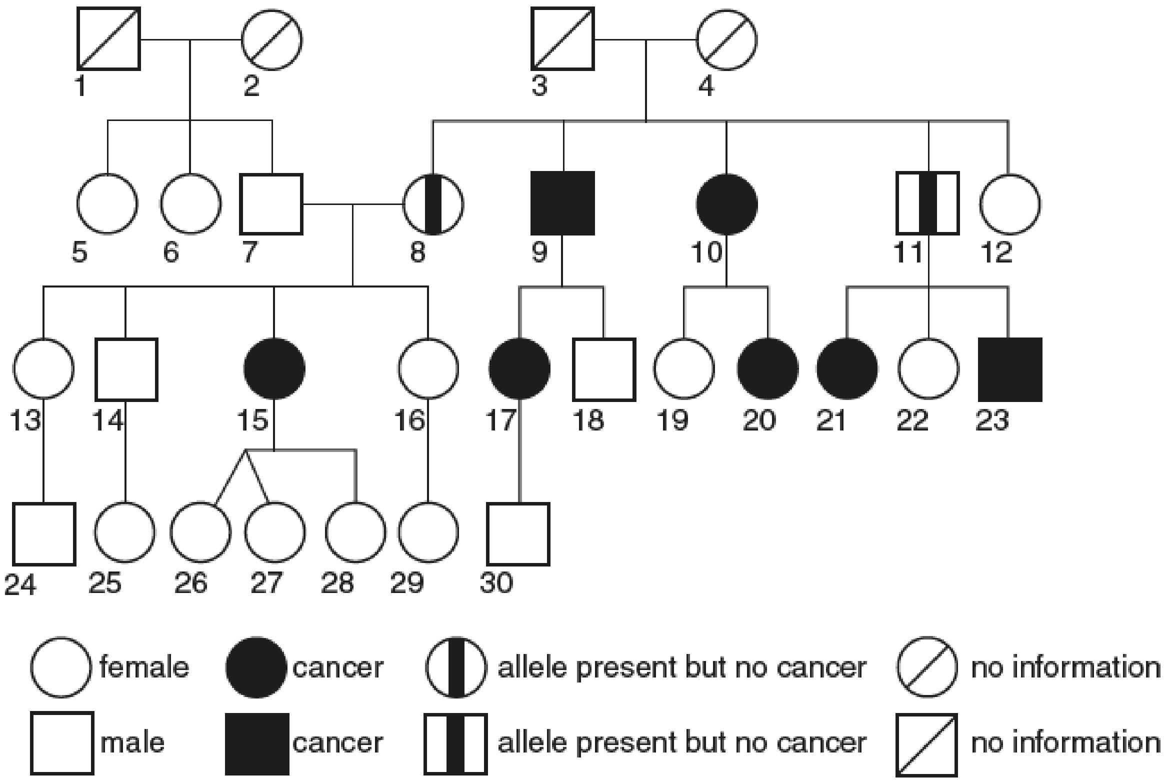

Fig. 2.1

(d) Discuss the extent to which Fig. 2.1 provides evidence that a faulty *BRCA2* allele increases the risk of a person developing cancer.

[4]

[Total: 13]

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BOOKLET C

Name: .... (   ) **CG:** ....

3 Tuberculosis (TB) is an infectious disease caused by a bacterial pathogen.

Fig. 3.1 shows the infection cycle of the pathogen.

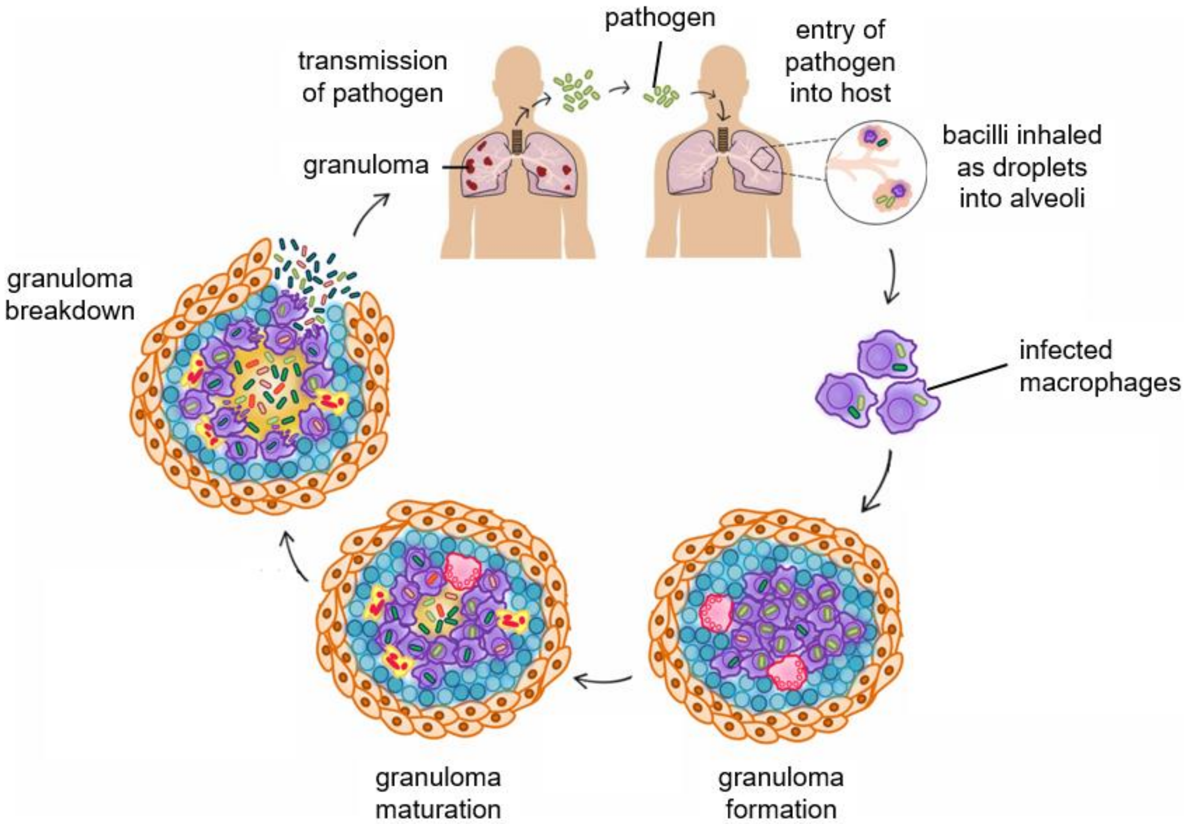

Fig. 3.1

(a) **(i)** Name the pathogen that causes tuberculosis.

.... [1]

(ii) With reference to Fig. 3.1, describe how latent TB is established in an individual upon infection.

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.... [2]

The global increase in drug-resistant tuberculosis has prompted a search for alternative therapies.

Researchers hypothesised that the use of a combination of antibiotics is useful in treatment of drug-resistant tuberculosis. In the investigation, researchers identified the minimum inhibitory concentration (MIC) of antibiotics against multiple strains of the pathogen. The antibiotics used in each experimental set-up are as follows:

- **A**: amoxicillin only
- **B**: a combination of amoxicillin and clavulanate (amoxicillin / clavulanate)
- **C**: meropenem only
- **D**: a combination of meropenem and clavulanate (meropenem / clavulanate)

Amoxicillin and meropenem are β-lactam antibiotics. Clavulanate is a β-lactamase inhibitor, inhibiting the action of β-lactamase that is secreted by antibiotic-resistant bacteria.

The MIC refers to the lowest concentration of antibiotic being tested that inhibits the growth of bacteria. A higher MIC indicates that the bacteria are resistant against the antibiotic, while a lower MIC indicates that the bacteria are more susceptible towards the antibiotic.

Fig. 3.2 shows the results of the investigation.

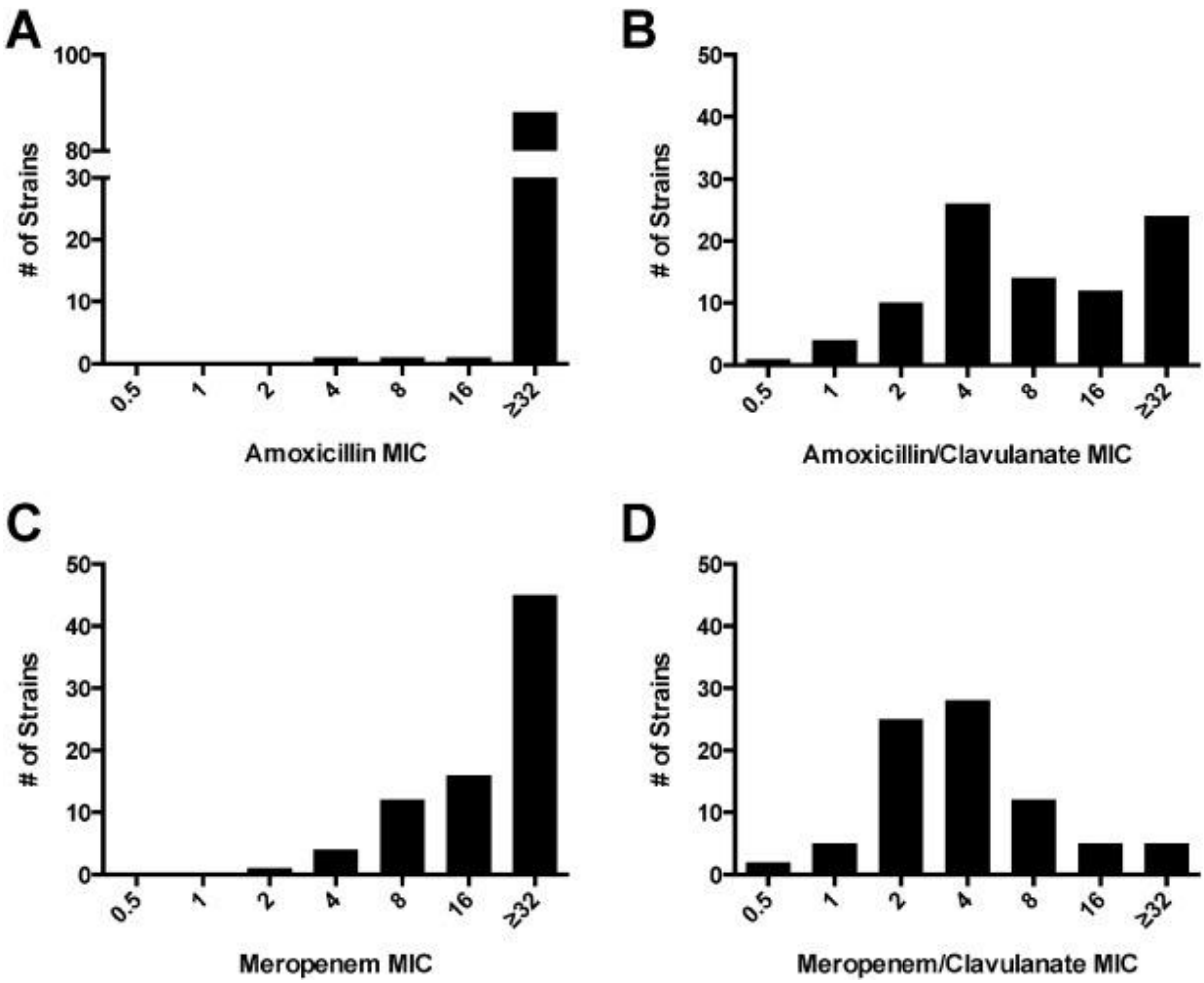

Fig. 3.2

(b) With reference to Fig. 3.2 and the information provided, discuss whether the results of the investigation support the researchers' hypothesis.

[3]

(c) Penicillin, like amoxicillin and meropenem, is a β-lactam antibiotic.

Describe the mode of action of penicillin on bacteria.

[2]

(d) Peptidoglycan and cellulose are important components of cell walls in bacteria and plants respectively.

Compare the structures of peptidoglycan and cellulose.

[2]

[Total: 10]

Section B

Answer **ONE** question in this section.

Write your answers on the lined paper provided at the end of this Question Paper.
Your answers should be illustrated by large, clearly labelled diagrams, were appropriate.
Your answers must be in continuous prose, where appropriate.
You answers must be set out in parts **(a)**, **(b)**, etc., as indicated in the question.

4 **(a)** Explain the various roles of membranes in cells. [12]

(b) Discuss how antibody diversity is generated in B cells, and how plasma cells are able to secrete specific antibodies against a particular antigen. [13]

5 **(a)** Using *lac* operon as an example, explain how gene regulation is achieved in prokaryotes. [12]

(b) Ions are atoms or molecules with a net electric charge.

Discuss the role of ions in ensuring the continuity of life. [13]

Question ____________